

المسألة = إيجاد

$$1) \int (x^2 - 2x + 5) dx = \frac{x^3}{3} - x^2 + 5x.$$

$$2) \int (x^3 + x)^5 (3x^2 + 1) dx = \frac{(x^3 + x)^6}{6}$$

$$\begin{aligned} 3) \int \sqrt{2x+1} dx &= \int (2x+1)^{\frac{1}{2}} = \frac{1}{2} \int (2) (2x+1)^{\frac{1}{2}} \\ &= \frac{1}{2} \left[\frac{(2x+1)^{\frac{3}{2}}}{\frac{3}{2}} \right] = \frac{1}{2} \cdot \frac{2}{3} (2x+1)^{\frac{3}{2}} \\ &= \frac{1}{3} (2x+1)^{\frac{3}{2}} \\ &= \frac{1}{3} \sqrt{(2x+1)^3} \end{aligned}$$

$$4) \int x^2 \cdot e^{x^3} dx = \frac{u = x^3}{du = 3x^2 dx}$$

$$\begin{aligned} \int \frac{1}{3} e^u du &= \frac{1}{3} \int e^u du = \frac{1}{3} e^u \\ &= \frac{1}{3} e^{x^3} + C \end{aligned}$$

$$\begin{aligned} 5) \int \frac{dx}{e^x + e^{-x}} &= \int \frac{e^x}{e^x(e^x + e^{-x})} dx = \int \frac{e^x}{e^{2x} + 1} dx \quad \text{Use } u = e^x \\ &= \int \frac{du}{u^2 + 1} = \arctan(u) \\ &= \arctan(e^x) \end{aligned}$$

$$6) \int \frac{2x \sqrt{x}}{\sqrt[3]{3x+1}} dx = \frac{2x^{\frac{3}{2}}}{(3x+1)^{\frac{1}{3}}}$$

$$u = 3x+1$$

$$x = \frac{u-1}{3}$$

$$\frac{dx}{du} = \frac{1}{3}$$

$$\int \frac{x \, dx}{\sqrt{x^2+1}} = \int (x) (x^2+1)^{-\frac{1}{2}} dx$$

$$= \frac{1}{2} \int (2x) (x^2+1)^{-\frac{1}{2}} dx$$

$$= \frac{1}{2} \int \frac{(x^2+1)^{-\frac{1}{2}}}{\frac{1}{2}} = (x^2+1)^{-\frac{1}{2}} = \sqrt{x^2+1} \, dx$$

$$u = x^2+1$$

$$du = 2x \, dx$$

$$\int \frac{du}{1+u^2} = \int \frac{du}{1+u^2} = \arctan u + C$$

$$\int u^{-\frac{1}{2}} \frac{du}{2}$$

$$\frac{1}{2} \int u^{-\frac{1}{2}} du$$

$$\frac{1}{2} \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{1}{2} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{1}{2} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$u = x^2+1$$

$$u = \frac{2}{3}x + 1$$

$$\Rightarrow du = \frac{2}{3} dx$$

$$dx = \frac{3}{2} du$$

$$I = \frac{5}{9} \int \frac{1}{1+u^2} \frac{3}{2} du$$

$$= \frac{5}{6} \int \frac{1}{1+u^2} du$$

$$= \frac{5}{6} \arctan(u) + C = \frac{5}{6} \arctan\left(\frac{2}{3}x + 1\right) + C$$

$$= \frac{5}{6} \arctan(u) + C$$

$$\int \sin^2(x) \cos^2(x) dx =$$

$$I = \int u^2 du \quad \begin{matrix} u = \sin x \\ du = \cos x dx \end{matrix}$$

$$= \frac{u^3}{3} + C = \frac{\sin^3 x}{3} + C$$

$$u = \tan x \\ du = \sec^2 x dx$$

$$I = \int u^2 du = \frac{u^3}{3} + C = \frac{\tan^3 x}{3} + C$$

$$\int x \cdot e^{x^2} dx \quad \begin{matrix} u = x^2 \\ du = 2x dx \end{matrix}$$

$$\int \frac{1}{2} e^u du = \frac{1}{2} e^u + C = \frac{1}{2} e^{x^2} + C$$

$$\int \frac{\ln x}{x} dx \quad \begin{matrix} u = \ln x \\ du = \frac{1}{x} dx \end{matrix}$$

$$\int u du = \frac{u^2}{2} + C = \frac{\ln^2 x}{2} + C$$

~~$\int (x^3+x)^5 (3x^2+1) dx$~~

$$\int (x^3+x)^5 (3x^2+1) dx$$

$$\begin{aligned}\int u^5 du &= \frac{u^6}{6} \\ &= \frac{(x^3+x)^6}{6}\end{aligned}$$

$$\begin{aligned}u &= x^3+x \\ du &= 3x^2+1\end{aligned}$$

$$\int \sqrt{2x+1} dx = \int (2x+1)^{\frac{1}{2}} dx$$

$$= \frac{1}{2} \int u^{\frac{1}{2}} du$$

$$\begin{aligned}u &= 2x+1 \\ du &= 2 dx\end{aligned}$$

$$= \frac{1}{2} \cdot \frac{u^{\frac{3}{2}}}{\frac{3}{2}}$$

$$= \frac{1}{3} u^{\frac{3}{2}} = \frac{1}{3} (2x+1)^{\frac{3}{2}}$$

$$\int \frac{2z}{\sqrt{z^2+1}} dz = \int 2z \cdot (z^2+1)^{-\frac{1}{2}}$$

$$= 2 (z^2+1)^{\frac{1}{2}} + C$$

$$= 2\sqrt{z^2+1} + C$$

$$\int x\sqrt{2x+1} dx =$$

$$\begin{aligned}u &= 2x+1 \Rightarrow x = \frac{u-1}{2} \\ du &= 2 dx \Rightarrow dx = \frac{du}{2}\end{aligned}$$

$$\int \frac{u-1}{2} \cdot \frac{1}{2} u^{\frac{1}{2}} \frac{du}{2} = \frac{1}{4} \int (u-1) u^{\frac{1}{2}} du$$

$$\begin{aligned}&= \frac{1}{4} \int u^{\frac{3}{2}} - u^{\frac{1}{2}} du \\ &= \frac{1}{4} \left(\frac{u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right) + C\end{aligned}$$

$$\int \sin^2 x \, dx.$$

$$\Rightarrow \int \frac{1 + \cos 2x}{2} \, dx$$

$$\int \frac{1}{2} + \frac{1}{2} \cos 2x \, dx$$

$$= \frac{1}{2}x + \frac{1}{4} \sin 2x + c$$

• anzahl = 16 [5]

$$\sin^2 x = \frac{1 + \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\int \cos^2 x \cdot x \, dx$$

$$= \int \left(\frac{1 + \cos 2x}{2} \right) \cdot x \, dx$$

$$= \frac{1}{2} \int (1 + \cos 2x) x \, dx$$

$$= \frac{1}{2} \int x + x \cos 2x$$

$$= \frac{1}{2} \int x + 2x \cos 2x$$

$$= \frac{1}{2} \int x + \frac{1}{2} \int$$

$$2 \sin x \cdot \cos x = \sin 2x$$

$$\int \sin x \cdot \cos x \, dx.$$

$$\int \frac{\sin 2x}{2} \, dx.$$

$$= \frac{1}{2} \int \sin 2x \, dx = \frac{1}{2} \left(-\frac{1}{2} \cos 2x \right) + c$$

$$\int \tan^2 x \, dx$$

$$= \int \sec^2 x - 1 \, dx$$

$$= \tan x - x + C$$

$$(\tan x)' = \sec^2 x$$

$$\sec^2 x = 1 + \tan^2 x$$

$$\int \sec^2 x = \tan x$$

$$\sec x \tan x = \sec x$$

$$\int \sec x \tan x = \sec x$$

النكاح بالتعويض :

نكاح مجموع = مجموع نكاحات

$$\int (x e^x) \, dx = \int x \, dx + \int e^x \, dx$$

نكاح x = نكاح e^x لا يلائم

$$\int x e^x \, dx = \int x \, dx + \int e^x \, dx$$

نكاح x = نكاح e^x لا يلائم

$$\int x e^x \, dx = \int x \, dx + \int e^x \, dx$$

$$= \int x e^x - \int e^x \, dx$$

$$= x e^x - e^x = e^x (x - 1) + C$$

$$\int (x+2)e^x dx$$

$$u = x+2$$

$$du = dx$$

$$v = e^x$$

$$= (x+2)(e^x) - \int e^x dx$$

$$= (x+2)e^x - e^x + C$$

$$= (e^x)(x+2-1) + C$$

: die ist es

$$\int x \sin x dx$$

$$u = x$$

$$du = dx$$

$$v = \sin x$$

$$= x \cos x - \int \cos x dx$$

$$= x \cos x + \sin x + C$$

$$\int x \cos x dx$$

$$u = x$$

$$du = dx$$

$$v = \cos x$$

$$= x \sin x - \int \sin x dx$$

$$= x \sin x + \cos x + C$$

Table of Integrals

Function $f(x)$

0
 Konstant k
 x^n
 x^2
 x^n
 $x^{-1} = \frac{1}{x}$
 $\sin x$
 $\sin kx$
 $\cos x$
 $\cos kx$
 $\sec kx$
 $\tan kx$
 e^n
 $-e^n$
 $\frac{e^n}{k}$
 $\cosh kx$
 $\sinh kx$
 $\frac{1}{x^2+a^2}$
 $\frac{1}{x^2-a^2}$
 $\frac{1}{\sqrt{x^2+a^2}}$
 $\frac{1}{\sqrt{x^2-a^2}}$
 $\frac{1}{a^2-x^2}$
 $\frac{1}{\sqrt{x^2+k}}$
 $\frac{1}{\sqrt{a^2-x^2}}$
 $a^k x$

Indefinite integral $\int f(x) dx$
 constant

$kx + c$
 $\frac{x^2}{2} + c$
 $\frac{x^3}{3} + c$
 $\frac{x^{n+1}}{n+1} + c$ $n \neq -1$
 $\ln |x| + c$
 $-\cos x + c$
 $-\frac{\cos kx}{k} + c$
 $\sin x + c$
 $\frac{\sin kx}{k} + c$
 $\frac{1}{k} \ln |\sec kx + \tan kx| + c$
 $\frac{1}{k} \ln |\sec kx| + c$
 $\frac{e^n}{k} + c$
 $-\frac{e^n}{k} + c$
 $\frac{e^{kx}}{k} + c$
 $\frac{1}{k} \sinh kx + c$
 $\frac{1}{k} \cosh kx + c$
 $\frac{1}{a} \tan^{-1} \frac{x}{a} + c$ $a > 0$
 $\frac{1}{2a} \ln \frac{x-a}{x+a} + c$ $|x| > a > 0$
 $\sinh^{-1} \left(\frac{x}{a} \right) + c$ $a > 0$
 $\cosh^{-1} \left(\frac{x}{a} \right) + c$ $x \geq a > 0$
 $\frac{1}{2a} \ln \frac{a+x}{a-x} + c$ $|x| < a$
 $\ln (x + \sqrt{x^2+k}) + c$
 $\sin^{-1} \left(\frac{x}{a} \right) + c$ $-a \leq x \leq a$
 $\frac{1}{k \ln a} a^{kx} + c$ $a > 0, a \neq 1$

Common functions and their derivatives

Function

Derivative

constant

0

x

1

kx

k

x^n

$n x^{n-1}$

kx^n

$k n x^{n-1}$

e^x

e^x

$k e^x$

$k e^{kx}$

e

$\ln x$

$\frac{1}{x}$

$\ln kx$

$\frac{k}{x}$

$\sin x$

$\cos x$

$\sin kx$

$k \cos kx$

$\sin(kx + \alpha)$

$k \cos(kx + \alpha)$

$\cos x$

$-\sin x$

$\cos kx$

$-k \sin kx$

$\cos(kx + \alpha)$

$-k \sin(kx + \alpha)$

$\tan x$

$\sec^2 x$

$\tan kx$

$k \sec^2 kx$

$\tan(kx + \alpha)$

$k \sec^2(kx + \alpha)$

$\sinh x$

$\cosh x$

$\sinh kx$

$k \cosh kx$

$\tanh x$

$\operatorname{sech} x$

$\tanh kx$

$k \operatorname{sech} kx$

$\sin^{-1} x$

$\frac{1}{\sqrt{1-x^2}}$

$\sin^{-1} kx$

$\frac{k}{\sqrt{1-k^2x^2}}$

$\cos^{-1} x$

$\frac{-1}{\sqrt{1-x^2}}$

$\cos^{-1} kx$

$\frac{-k}{\sqrt{1-k^2x^2}}$

$\tan^{-1} x$

$\frac{1}{1+x^2}$

$\tan^{-1} kx$

$\frac{k}{1+k^2x^2}$

$$I_2 = \int x^2 \cos x \, dx \quad \begin{array}{l} u = x^2 \\ du = 2x \, dx \end{array} \quad \begin{array}{l} dv = \cos x \, dx \\ v = \sin x \end{array}$$

$$I_2 = x^2 \sin x - \int 2x \sin x \, dx \quad \begin{array}{l} u = 2x \\ du = 2 \, dx \end{array} \quad \begin{array}{l} dv = \sin x \, dx \\ v = -\cos x \end{array}$$

$$I_2 = x^2 \sin x - \int -2x \cos x - \int -2 \cos x \, dx$$

$$= x^2 \sin x + 2x \cos x - 2 \sin x + C$$

$$= x^2 \sin x + 2(x \cos x - \sin x) + C$$

$$\int x^2 \sin x \, dx \quad \begin{array}{l} u = x^2 \\ du = 2x \, dx \end{array} \quad \begin{array}{l} dv = \sin x \, dx \\ v = -\cos x \end{array}$$

$$I_3 = x^2 \cos x - \int -2x \cos x \, dx$$

$$= x^2 \cos x + 2 \int x \cos x \, dx \quad \begin{array}{l} u = x \\ du = dx \end{array} \quad \begin{array}{l} dv = \cos x \, dx \\ v = \sin x \end{array}$$